

Master Thesis Presentation

- Hello everyone. Today, I will be presenting what I have been working on over the past few months.
- I developed a surface grinding controller for a seven-degree-of-freedom Franka robot. The control method I used is Cartesian impedance control.

Motivation

- The main problem in surface grinding is that we cannot ensure that the configured and physical surfaces are perfectly aligned. Calibration leaves some uncertainty in the surface orientation.
- Joint friction can also prevent the tool from reaching the desired orientation. The tool may therefore first contact the surface at an edge.
- The idea is to use impedance control to make rotation compliant. Pressing the tilted tool creates a contact moment, which can rotate the tool towards alignment.

Contact demonstration

- The video shows pickup, approach, contact, and the hold before grinding.
- During the hold, I change the surface orientation. Normal pressing continues as contact moments rotate the tool.
- The desired orientation stays fixed. This is a qualitative demonstration of compliant contact.

Overview

- After this introduction, I explain the Cartesian controller, followed by the contact experiments and results.
- The separate null-space controller, experiment and results follow together. I finish with the conclusion and future work.

Cartesian pose

- To begin, let's first introduce Cartesian pose. Cartesian pose describes the position and orientation of the gripper, also called the end effector, or EE.
- It has three positional degrees of freedom and three rotational degrees of freedom. The pose is determined through forward kinematics from the positions of all seven joints.

Surface frame

- To choose compliance relative to the surface, we use a surface frame. The normal points out of the surface, while the two perpendicular tangents lie in the surface.
- This lets us choose different stiffnesses for pressing, tangential motion and rotation. I will use these directions when explaining the contact settings.

Cartesian impedance controller

- Impedance control uses a spring-damper model that produces a force to correct position and a moment to correct rotation. Together, the force and moment form what is called a wrench.
- K and D are the stiffness and damping matrices. In the surface frame they are diagonal, so translation and rotation are decoupled in this form.

Real-time control

- The controller compares the reference state with the measured state. Here, state includes both pose and velocity, so stiffness and damping act on their respective errors.
- The wrench is then mapped to joint torques through the transpose of the Jacobian. The Jacobian itself maps the seven joint velocities to the linear and angular velocity of the end effector.

$$\tau = J^T(q)F$$

$$\begin{bmatrix} \dot{p}_{EE} \\ \omega_{EE} \end{bmatrix} = J(q)\dot{q}$$

- The complete command also includes model compensation and any enabled null-space torques. The controller operates in real time, with an update every millisecond.

Centre of compliance (CoC)

- An interesting idea presented in this work is to couple force and moment so that a pressing force also produces a rotational moment. We do this by choosing a virtual reference point called the centre of compliance, or CoC.
- We can illustrate this behaviour here. In the left panel, the displacement from the tool centre point gives an additional moment that supports alignment.
- In the right panel, the displacement is on the other side and the additional moment opposes alignment. This motivates the reference-point transformation on the next slide.

Translation–rotation coupling

- The tool centre point, or TCP, is the controlled reference point on the end effector. We shift the reference point of the impedance model to the virtual CoC, with the displacement defined from the TCP to that point.
- The commanded force then acts as if applied at the virtual point. This changes the controller's force and moment response without moving the physical contact point.
- For the local, small-rotation model, we transform the stiffness and damping matrices using the point-shift adjoint. The resulting matrices contain off-diagonal terms, which couple translation and rotation.

Contact experiment

- Our contact experiment proceeds as follows. First, we orient the tool towards the configured surface and slowly approach it along its normal.
- During contact establishment, compliant rotation allows the contact moment to turn the tool towards alignment. Grinding then begins along the tangential direction t_2 .
- For translation, we choose a compliant normal stiffness to allow pressing against the surface. The tangential directions remain stiff to limit unwanted motion during alignment.
- For rotation, the normal direction is stiff, while rotations about the tangential directions are compliant. This allows the tool to tilt in response to contact moments.

Plausibility experiment

- Before looking at the contact results, we first check whether the controller produces the expected force and moment. This video introduces the plausibility experiment.
- The next two plots compare the commanded and model-estimated responses with the spring-law predictions for manual translation and rotation.

Normal-force plausibility assessment

- First, we manually push the end effector about 20 mm along the surface normal, while keeping the desired pose fixed and the CoC at the TCP. With a normal stiffness of 1000 newtons per metre, the measured position-error change predicts a force of about minus 19.65 newtons.

$$F_{n,qs} = (1000 \text{ N/m})(-0.019650 \text{ m}) = -19.650 \text{ N}$$

- In the shaded stationary interval, the mean commanded force is minus 19.64 newtons, very close to the quasi-static prediction. The model-estimated force is minus 22.29 newtons, about 13.5 percent different from the command.
- The commanded response therefore agrees with the spring law in this test. The model estimate provides a separate plausibility check.

Moment plausibility assessment

- We do the same for the moment by manually rotating the tool about the first surface tangent while retaining the normal push. The measured rotational-error change is about 6.57 degrees relative to that loaded baseline.
- With a rotational stiffness of 15 newton metres per radian, the prediction is about 1.72 newton metres. The stationary commanded moment matches this value, while the model estimate is about 1.85 newton metres, a difference of roughly 8 percent.

$$M_{t_1,qs} = (15 \text{ N m/rad}) \left(6.565 \frac{\pi}{180} \text{ rad} \right) \approx 1.719 \text{ N m}$$

- Together, these two checks support the expected force and moment response for the tested errors.

Commanded and estimated wrench

- Next, we compare the commanded and model-estimated wrench during a separate contact trial, with the CoC at the TCP. The upper plot shows normal force and the lower plot shows moment about the first surface tangent.
- The force curves follow each other closely as pressing increases. During alignment, the estimated moment is much larger than the commanded rotational moment.
- Later, the force and moment settle and the commanded and estimated values become close. The moment difference during alignment is consistent with an external contact moment driving the compliant rotation.

Effect of rotational stiffness

- For the contact results, angular error describes the tool-normal tilt relative to the calibrated surface, about the first surface tangent. It is calculated from measured end-effector orientation, and a smaller magnitude means less error about this tangent.
- This plot shows that higher rotational stiffness produces a larger remaining angular error. Increasing the stiffness from 5 to 50 newton metres per radian increases the error from about 1.75 to 6.58 degrees.
- The rotational spring therefore resists the contact-driven alignment more strongly. These points are three-trial means, with error bars showing one sample standard deviation.

Effect of CoC position

- Next, this plot shows how the centre of compliance affects the angular error. We moved it from minus 100 to plus 100 mm along the second surface tangent for positive and negative entry tilts.
- For the positive entry tilt, shown in black, positive displacement supports alignment and reduces the error. Large negative displacement opposes it and leaves a much larger error.
- For the negative entry tilt, shown in blue, the smallest tested error magnitude occurs near plus 10 mm. At larger positive displacements the error becomes negative and its magnitude grows.
- The important point is that the effect depends on the entry direction and the chosen displacement. A larger displacement does not automatically give a smaller angular error.

Angular error and interaction wrench

- Next, we compare representative time histories at CoC displacements of minus 40 mm, zero and plus 40 mm. The three panels show angular error, model-estimated normal force and model-estimated TCP moment.
- For this positive entry tilt, the supporting plus-40-mm setting reaches its final-error band at about 2.5 seconds, compared with 3.6 seconds at the TCP. The band means staying within 0.1 degree of each trace's own final value.
- The opposing minus-40-mm setting leaves the angular error close to its entry value. Its estimated moment continues to build while little alignment occurs.

- The force traces are broadly similar and settle near minus 80 newtons, while the moment traces differ clearly. The supporting displacement changes the moment response and speeds up the angular response in this comparison.

Null-space controller

- Another study in this work analysed null-space behaviour. Our Cartesian pose has six degrees of freedom, while the robot has seven joints, leaving one redundant direction when the Jacobian has full rank. Several joints can move together in this direction without changing the instantaneous Cartesian motion.
- We introduced a damping torque to oppose this motion.
- The conditioning torque seeks a configuration with a larger minimum singular value of the Jacobian. At a singularity, the Jacobian loses rank and the end effector loses an instantaneous motion direction. Near a singularity, some motions require very high joint velocities.

Disturbance - Without null-space control to damping

- This first video shows the response to a torque disturbance, first without null-space control and then with damping. It gives a qualitative view of how the joints respond.

Disturbance - Conditioning only

- This is the continuation of the same demonstration, now with conditioning only. The following experiment gives the quantitative comparison between the null-space settings.

Null-space experiment

- To study this case, we held the captured Cartesian pose as the reference and added a commanded torque disturbance equivalent to a virtual force of 20 newtons on link three. Through the Jacobian mapping, this disturbance acted mainly on joint one.
- With the disturbance applied, we compare four conditions with three trials each: no null-space control torque, conditioning alone, damping alone, and both torques together. The enabled settings are 2 newton metres for conditioning and 2 newton metre seconds per radian for damping.
- The following plots show the same four-second interval after disturbance onset. The curves are three-trial means, and the shading shows one sample standard deviation.

Cumulative joint motion

- Cumulative joint motion accumulates the magnitude of projected motion using all seven joint velocities. It includes movement in both directions, so reversals do not cancel each other.
- With damping alone, cumulative motion is about 25 percent lower than without null-space torque. Conditioning reduces it further in this comparison.
- Adding damping to conditioning reduces cumulative motion by about another 51 percent compared with conditioning alone. The combined setting gives the smallest cumulative motion among these four conditions.

Jacobian conditioning

- Next, we look at the minimum singular value of the Jacobian. Without null-space torque, and with damping alone, this indicator decreases during the disturbance.
- Conditioning alone and combined control keep it nearly constant. The small difference between these two curves does not establish a meaningful conditioning advantage.

Joint motion over time

- Finally, we look at the measured angle change of joint one, which receives most of the disturbance torque. Damping reduces this motion, and the conditioning and combined settings keep it much smaller.
- Both panels show the full four seconds, with a larger vertical scale on the right to reveal the small motions. Conditioning alone produces repeated reversals, and adding damping reduces total motion while reversals remain.

Conclusion

- To conclude, compliant rotation allowed contact moments to reduce angular error. Higher rotational stiffness left a larger error, and the centre of compliance could either support or oppose the contact moment.
- The null-space controller reduced redundant joint motion under disturbance. Combining conditioning with damping reduced cumulative motion further than conditioning alone in the main comparison.
- Future work could use a more rigid tool mount for better measurements. The CoC could also be adapted to support alignment and then returned to the TCP.
- Thank you for your attention.

Opposing moment

- This video illustrates the opposing-moment case. An additional moment that opposes the contact-driven rotation can hinder alignment.
- It complements the supporting and opposing configurations in the earlier illustration. The contact plots provide the quantitative comparison.

Null-space torques and projector

- The kinematic projector selects null-space torque directions. It uses the Jacobian pseudoinverse J^+ and the seven-by-seven identity matrix.

$$N_{\tau} = I_7 - J^{\top} J^{+ \top}$$

- Damping and conditioning add to give the total null-space torque.

$$\tau_{\text{null}} = \tau_d + \tau_{\sigma}$$

- Damping opposes projected joint velocity and is zero at rest. The damping coefficient sets its strength.

$$\tau_d = -d_{\text{null}} N_\tau \dot{q}$$

- Conditioning probes both directions along the unit null-space vector and chooses the direction with the larger minimum singular value. The selector is plus one or minus one, and zero within the deadband.

$$\tau_\sigma = k_\sigma N_\tau (s_\sigma v_7)$$

- The conditioning coefficient sets the torque magnitude. Unlike damping, this torque can be nonzero at rest.

Angular quantities

- The measured angular offset is the tool-normal tilt at contact entry relative to the calibrated surface. The angular error describes that tilt during contact, resolved about the first surface tangent.
- The red arc shows the entry offset and the blue arc shows the remaining error at contact end. Both are calculated from measured end-effector orientation and the calibrated tool normal.
- This is different from the controller's orientation error relative to its held reference. A zero first-tangent component alone does not establish perfect physical alignment, because other components and calibration or mount uncertainty remain.

Baseline angular error

- This plot shows the baseline with the CoC at the TCP. The measured entry offsets are about 0.69, plus 9.31 and minus 9.41 degrees.
- The corresponding remaining angular-error components are about 1.65, 1.75 and 1.41 degrees. The two larger entry tilts are reduced substantially, while the nominal-zero case retains a small error.
- The response is not perfectly symmetric, and friction, calibration and tool-mount effects may contribute. This plot alone does not separate those contributions.

Sources of angular offset

- This sketch separates two sources of angular offset at surface entry. The physical surface can differ from the configured surface, and the actual tool orientation can differ from the desired orientation.
- The dashed black line shows the desired orientation, while the green line shows the tool face. Together, these differences explain why the tool can meet the physical surface at a tilt.